

UMA MEMORY BUS CONTENTION & THE PRE-INFERENCE INGESTION TAX

Hardware bus collision between unthrottled camera DMA, NPU tensor streaming, and CPU real-time threads

EMBEDDED APPLICATION PROCESSOR (SoC) — UNIFIED MEMORY ARCHITECTURE (UMA)

MASTER 1: SENSOR INGESTION (ISP/DMA)

- MIPI CSI-2 uncompressed 4K stream
- Ingestion rate: **1.49 GB/s** burst writes
- Unregulated DMA floods memory queues

MASTER 2: VISION-LANGUAGE NPU

- ViT patch tokenization + VLM weights
- Memory demand: **24.0 GB/s** peak read
- KV-cache streaming stalls on bus delay

MASTER 3: CPU CORES

- IPC lease coordination
- Real-time threads: **2.5 GB/s**
- Suffers priority inversion

SHARED BUS ARBITER



UMA Bottleneck:

- Shared LPDDR5 Channel
- Peak BW: 34 GB/s Limit
- Unmanaged QoS = Starvation

SHARED LPDDR5 DRAM (8-16 GB)

Camera Framebuffers (DMA)

VLM Weights & KV Cache

Zero-Copy Ring Buffer

Linux OS Kernel & Stack

✗ NAIVE APPROACH: MEMCPY & UNMANAGED BUS BURSTS

Nominal Latency (P50):

22.4 ms

Tail Latency (P99):

135.8 ms ⚠️

- **3x User/Kernel Space Copies:** V4L2 kernel buffer → OpenCV Mat → Tensor tensor.
- **Bus Starvation:** Camera bursts lock out NPU weight streaming during critical tokens.
- **Causal Consequence:** Breaches Freshness Wall ($\Delta t_{fresh} > 60\text{ ms}$), triggering motor aborts.

✓ PHYSICAL AI CONTRACT: ZERO-COPY DMA + HARDWARE QoS

Nominal Latency (P50):

5.8 ms

Tail Latency (P99):

8.2 ms (Deterministic)

- **Zero-Copy DMABUF Pipeline:** ISP writes directly to memory-mapped tensor buffer.
- **Hardware QoS Slots:** Strict arbiter bandwidth caps guarantee real-time CPU/NPU access.
- **Bounded Metrology:** Bounded $P99 \leq 8.2\text{ ms}$ preserves metric state estimation freshness.